Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course
Computer Graphics- Core
Sem. V– Program
INFT 2026-2027– Odd Semester
Faculty – Dr. Santosh P. Tamboli Cluster Mentor) Dr.Sushopti Gawade

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library is member of IIT Bombay Library	Watch former lectures captured in LMS at VIT

1.a **Course Objectives (write in detail – as per NBA guidelines)**

Cognitive	What do you want students to know?	Students must know the fundamentals of database management systems, transactions, concurrency, distributed databases and data warehousing.
Affective	What do you want students to think / care about?	Students must care about database modelling, choice of ETL tools and database security concerns.
Behavioral	What do you want students to be able to do?	Students must be able to design databases, distributed databases and organize large amounts of data in a data warehouse.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if what you understand is correct. The v-Refer Link for this course:

The v-Refer Link:

https://vrefer.vidyalankar.edu.in/index.php/apps/files/files/920871?dir=/vit%20vrefer/INFT/SEM%20V/2026-27/Verticals/PC_PCC/CG%20%28PCIT03T%26P%29/ST

Creation of microsite or teams link:-

https://teams.cloud.microsoft/l/team/19%3AWcD7ivE4NFSYmF4j-ZXOJXvgC3hUeCACTnXAUFiT7kk1%40thread.tacv2/conversations?groupId=80824af0-6353-4f02-9001-8097f0525f1e&tenantId=c7b00d7f-ad99-442a-b12f-c2c912044fdc

Team code: jwwfmr8

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the

internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Explain the core concepts of computer graphics including coordinate systems and graphics primitives.	Module 1
CO2	Apply scan conversion and polygon filling techniques to generate and manipulate basic graphical primitives.	Module 2
CO3	Apply the concepts of 2D and 3D transformations	Module 3
CO4	Analyze visible surface detection algorithms and implement illumination and shading models for realistic rendering.	Module 4
CO5	Utilize and analyze viewing, clipping, and projection techniques to display 2D and 3D scenes accurately.	Module 5
CO6	Design and evaluate graphics systems and algorithms involving curves, surfaces, and display architectures.	Module 6

1.c Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

	Mapped to Learning Outcomes
CO1	*LO1.1 Explain the evolution, applications, and importance of computer graphics in various domains. 1.4.1* *LO1.2 Describe different coordinate systems used in computer graphics. 1.4.2* *LO1.3 Differentiate between raster scan and random scan display systems. 1.4.2* *LO1.4 Explain the architecture of raster graphics systems and display processors. 1.4.2* *LO1.5 Illustrate graphics primitives and their representation in computer graphics. 2.3.2* *LO1.6 Explain rendering and rasterization processes used in graphics systems. 1.4.2*
CO2	*LO2.1 Implement DDA algorithm for line drawing. 5.2.3* *LO2.2 Implement Bresenham's line drawing algorithm. 5.2.3* *LO2.3 Implement Bresenham's and Midpoint circle drawing algorithms. 5.2.3* *LO2.4 Compare aliasing and anti-aliasing techniques. 2.4.3* *LO2.5 Apply Boundary Fill and Flood Fill algorithms for polygon filling. 5.2.3* *LO2.6 Implement Scan Line polygon filling algorithm for graphical applications. 5.2.3*
CO3	*LO3.1 Apply translation, scaling, and rotation transformations on 2D objects. 3.4.3* *LO3.2 Perform composite transformations using homogeneous coordinates. 3.4.3* *LO3.3 Apply 3D geometric transformations on graphical objects. 3.4.6* *LO3.4 Perform rotation about an arbitrary axis in 3D space. 3.4.6* *LO3.5 Apply parallel and perspective projection techniques. 3.4.3* *LO3.6 Implement transformation matrices for real-world graphical applications. 5.2.3*
CO4	*LO4.1 Explain the need for visible surface detection in computer graphics. 1.4.2* *LO4.2 Compare Depth Buffer, Scan Line, and Area Subdivision methods. 2.4.4* *LO4.3 Analyze the performance of different hidden surface removal algorithms. 2.4.4* *LO4.4 Explain illumination models including diffuse and specular reflection. 1.4.2* *LO4.5 Implement Gouraud, Phong, and Constant shading techniques. 5.2.3* *LO4.6 Evaluate rendering quality using different shading techniques. 4.3.5*

CO5	*LO5.1 Explain the viewing transformation pipeline. 1.4.2* *LO5.2 Apply window-to-viewport transformation. 3.4.3* *LO5.3 Implement Cohen–Sutherland line clipping algorithm. 5.2.3* *LO5.4 Apply Liang–Barsky clipping algorithm for line clipping. 5.2.3* *LO5.5 Implement Sutherland–Hodgeman polygon clipping algorithm. 5.2.3* *LO5.6 Analyze different clipping techniques based on efficiency and applications. 2.4.4*
CO6	*LO6.1 Generate Bezier curves using control points. 3.4.6* *LO6.2 Construct B-Spline curves for geometric modeling. 3.4.6* *LO6.3 Generate fractal curves such as Koch and Hilbert curves. 3.4.6* *LO6.4 Design graphical models using curve representation techniques. 3.4.6* *LO6.5 Evaluate different curve generation techniques for graphics applications. 4.3.5* *LO6.6 Develop graphics applications integrating curves, transformations, and rendering techniques. 5.2.3*

1.d **Mapping of COs with POs (mark S: Strong, M: Moderate, W: Weak, Dash '–': not mapped) (List of POs is available in V-refer)**

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Life long learning
CO 1	5/36	1/36			1/36	1/36						1/36
CO 2	1/36	2/36	1/36		5/36							
CO 3	1/36	2/36	4/36		2/36							1/36
CO 4	2/36	3/36	1/36	2/36	2/36							1/36
CO5	1/36	3/36	2/36	1/36	4/36							1/36
CO6	1/36	2/36	4/36	2/36	3/36							1/36

1.e **Mapping of COs with PSOs (mark S: Strong, M: Moderate, W: Weak, Dash '–':not mapped)**

	PSO 1	PSO 2	PSO 3
CO 1	5/36	1/36	1/36
CO 2	2/36	4/36	
CO 3	2/36	4/36	
CO 4	3/36	3/36	1/36
CO 5	2/36	4/36	
CO6	2/36	3/36	1/36

1.f **Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course**

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning

Tick suitable category		✓					

Subject Code	Subject Name	Teaching Scheme			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
PCIT03T	Computer Graphics	2	-	-	2	-	-	2
PCIT03P	Computer Graphics Lab	-	2	-	-	1	-	1

Subject Code	Subject Name	Examination Scheme							
		Theory			Total (Theory)	Practical			Total (Practical)
		ISA	MSE	ESE		ISA	ESE	ORAL	
PCIT03T	Computer Graphics	15	20	40	75	--	--	--	--
PCIT03P	Computer Graphics Lab	--	--	--	--	25	--	25	50

Subject Code	Subject Name	MSE-1*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT03T	Big Data Analytics	Q.1	Module 1 30% Module 2 70%	Remember Understand Apply
		Q.2		

Subject Code	Subject Name	MSE-2*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT03T	Big Data Analytics	Q.1	Module 3 50% Module 4 50%	Remember Understand Apply
		Q.2		

Subject Code	Subject Name	ESE#		
		Q, No	Distribution	Relevant to Bloom Taxonomy
PCIT03T	Big Data Analytics	Q.1	Module 1 10% Module 2 10% Module 3 15% Module 4 15% Module 5 25% Module 6 25%	Remember Understand Apply
		Q.2		
		Q.3		
		Q.4		

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30

marks from thought provoking

1.g **Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course**

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	2 (ST)	ST	AVL	DG	ST	--	--	--	--
B	2 (ST)	ST	AVL	ST	AVL	--	--	--	--
C	2 (ST)	DG	DG	ST	DG	--	--	--	--

1.h **Office Hours (Faculty will be available in office in this duration for solving students' query)**

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Friday	4-5pm	CC2
B	Tuesday	4-5pm	CC2
C	Wednesday	4-5pm	CC2

2.a **Syllabus : Module Wise Teaching Hours and % Weightage in University Question Paper**

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Introduction and Overview of Graphics System. **Contents:** Definition and Representative uses of Computer Graphics, Classification of application areas, Overview of Coordinate Systems, Definition of Scan Conversion, Rasterization and Rendering. Raster Scan & Random Scan Displays, Architecture of Raster Graphics System with display processor, Architecture of Random Scan Systems.	2	10	30 (MSE-1)	05	PI 1.4.2 Understanding
Learning Outcome-1.1	LO1.1 Explain the evolution, applications, and importance of computer graphics in various domains. 1.4.1 LO1.2 Describe different coordinate systems used in computer graphics. 1.4.2 LO1.3 Differentiate between raster scan and random scan display systems. 1.4.2 LO1.4 Explain the architecture of raster graphics systems and display processors. 1.4.2 LO1.5 Illustrate graphics primitives and their representation in computer graphics. 2.3.2 LO1.6 Explain rendering and rasterization processes used in graphics systems. 1.4.2					
Learning Outcome-1.2						
Learning Outcome-1.3						
Learning Outcome-1.4						
Learning Outcome-1.5						

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-1.5						
Learning Outcome-1.6						
2	Raster Algorithms. **Contents:** Scan Conversions of Point, Line, and Circle: DDA Algorithm and Bresenham Algorithm for Line Drawing, Aliasing, Antialiasing Techniques like Pre and Post Filtering, Super Sampling, and Pixel Phasing. Filled Area Primitives: Scanline Polygon Fill Algorithm, Inside Outside Tests, Boundary Fill and Flood fill Algorithm	8	20	70 (MSE-1)	20	PI 1.4.2 Understanding PI 2.2.4 Analyzing
Learning Outcome-2.1	LO2.1 Implement DDA algorithm for line drawing. 5.2.3 LO2.2 Implement Bresenham's line drawing algorithm. 5.2.3 LO2.3 Implement Bresenham's and Midpoint circle drawing algorithms. 5.2.3 LO2.4 Compare aliasing and anti-aliasing techniques.2.4.3 LO2.5 Apply Boundary Fill and Flood Fill algorithms for polygon filling. 5.2.3 LO2.6 Implement Scan Line polygon filling algorithm for graphical applications. 5.2.3					
Learning Outcome-2.2						
Learning Outcome-2.3						
Learning Outcome-2.4						
Learning Outcome-2.5						
Learning Outcome-2.6						
3	Two Dimensional Geometric Transformations, Viewing and Clipping **Contents** Basic transformations: Translation, Scaling, Rotation. Matrix Representation and Homogeneous Coordinates, Composite Transformation. Viewing Transformation Pipeline and Window to Viewport Coordinate Transformation. Clipping Operations: Point Clipping, Line Clipping Algorithms: Cohen–Sutherland, Midpoint Subdivision, Liang–Barsky, Polygon Clipping Algorithms: Sutherland–Hodgeman and Weiler – Atherton Algorithm	8	20	50 (MSE-2)	30	PI 1.4.1 Remembering PI 1.4.2 Understanding
Learning Outcome-3.1	LO3.1 Apply translation, scaling, and rotation transformations on 2D objects.3.4.3 LO3.2 Perform composite transformations using homogeneous coordinates. 3.4.3 LO3.3 Apply 3D geometric transformations on graphical objects. 3.4.6 LO3.4 Perform rotation about an arbitrary axis in 3D space. 3.4.6 LO3.5 Apply parallel and perspective projection techniques. 3.4.3 LO3.6 Implement transformation matrices for real-world graphical applications. 5.2.3					
Learning Outcome-3.2						
Learning Outcome-3.3						
Learning Outcome-3.4						
Learning Outcome-3.5						

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-3.6						
4	Three-Dimensional Object Representations, Geometric Transformations and 3D Viewing **Contents:** Boundary Representation and Space Partitioning Representation: Polygon Surfaces, Bezier Curve B-Spline Curve, Fractal-Geometry: Fractal Dimension, Koch Curve. 3D-Transformations: Translation, Rotation, Scaling and Reflection. Composite Transformations: Rotation about an Arbitrary Axis. 3D-Transformation Pipeline Projections– Parallel and Perspective Projection and their types.	6	20	50 (MSE-2)	20	PI 2.1.2 Understanding
Learning Outcome-4.1 Learning Outcome-4.2 Learning Outcome-4.3 Learning Outcome-4.4 Learning Outcome-4.5 Learning Outcome-4.6	LO4.1 Explain the need for visible surface detection in computer graphics. 1.4.2 LO4.2 Compare Depth Buffer, Scan Line, and Area Subdivision methods. 2.4.4 LO4.3 Analyze the performance of different hidden surface removal algorithms. 2.4.4 LO4.4 Explain illumination models including diffuse and specular reflection. 1.4.2 LO4.5 Implement Gouraud, Phong, and Constant shading techniques. 5.2.3 LO4.6 Evaluate rendering quality using different shading techniques. 4.3.5					
5	Visible Surface Detection **Contents:** Classification of Visible Surface Detection Algorithm, Back Surface Detection Method: Depth Buffer Method, Scan Line Method, Area Subdivision Method.	4	20	-	15	PI 2.3.4 Analyzing
Learning Outcome-5.1 Learning Outcome-5.2 Learning Outcome-5.3 Learning Outcome-5.4 Learning Outcome-5.5 Learning Outcome-5.6	LO5.1 Explain the viewing transformation pipeline. 1.4.2 LO5.2 Apply window-to-viewport transformation. 3.4.3 LO5.3 Implement Cohen–Sutherland line clipping algorithm. 5.2.3 LO5.4 Apply Liang–Barsky clipping algorithm for line clipping. 5.2.3 LO5.5 Implement Sutherland–Hodgeman polygon clipping algorithm. 5.2.3 LO5.6 Analyze different clipping techniques based on efficiency and applications. 2.4.4					
6	Illumination Models and Surface Rendering **Contents:** Basic Illumination Models: Diffused reflection, Phong Specular Reflection Model, Halftone and Dithering Techniques, Polygon Rendering: Constant shading, Gouraud Shading, Phong Shading.	2	10	-	10	PI 2.1.2 Understanding

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-6.1 Learning Outcome-6.2 Learning Outcome-6.3 Learning Outcome-6.4 Learning Outcome-6.5 Learning Outcome-6.6	LO6.1 Generate Bezier curves using control points. 3.4.6 LO6.2 Construct B-Spline curves for geometric modeling. 3.4.6 LO6.3 Generate fractal curves such as Koch and Hilbert curves. 3.4.6 LO6.4 Design graphical models using curve representation techniques. 3.4.6 LO6.5 Evaluate different curve generation techniques for graphics applications. 4.3.5 LO6.6 Develop graphics applications integrating curves, transformations, and rendering techniques. 5.2.3					
	Total	**30**	**100**	**100**	**100**	

2.b **Prerequisite Courses**

No.	Semester	Name of the Course	Topic/s
1	1,2	Applied Mathematics	Matrices
2	2	VSEC01T (Structured Programming)	C Programming concepts

2.c **Relevance to Future Courses**

No.	Semester	Name of the Course
1	7	HMIT01 **(**UX Design, Evaluation and ARVR)

2.d **See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video**

Real Life Scenario	Concept Used
Special effects used in Entertainment Industry , Animation	All the concepts in course contributes
Computer Aided Design and Computer Aided Manufacturing (CAD / CAM)	All the concepts in course contributes
Virtual Reality , Graphical User Interface (GUI), Movie Production and Visual Effects (VFX)	All the concepts in course contributes

Graphical way of presenting the statistics in Corporates, Animation	All the concepts in course contributes

3. Past Results – Division-Wise and Topic-Wise Result Based Analysis

Details	Target –DEC 2026	DEC 2025	MAY 2025	DEC 2024
Course Passing % – Average of 2 Divisions	98	97	NA	NA
Marks Obtained by Course Topper (mark/100)	98	96	NA	NA

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
May 2024	NA	NA	NA	NA	NA	NA
May 2023	NA	NA	NA	NA	NA	NA
May 2022	NA	NA	NA	NA	NA	NA

Topics which affect results negatively	Module Number	Recommendations to overcome these issues & improve result in future
2D and 3D Transformations	4	Solve more numerical on 2D and 3D Transformations

4 All the Learning Resources – Books and E-Resources

4.a List of Text Books (T – Symbol for Text Books) to be Referred by Students

Sr. No	Text Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Computer Graphics	S. Harrington	McGraw-Hill Publications	2nd Edition	All	Yes
2	Computer Graphics with OpenGL	Donald D. Hearn and Baker	Mc Graw Hill	4th Edition	All	Yes
3	Procedural Elements for Computer Graphics	D. Rogers	McGraw-Hill Publications	2nd Edition	ALL	Yes
4	Computer Graphics with Virtual Reality	R. K Maurya	Pearson	2nd	ALL	Yes

4.b **List of Reference Books (R – Symbol for Reference Books) to be Referred by Students**

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Computer Graphics Principles and Practice	J. Foley, V. Dam, S. Feiner, J. Hughes	Pearson Education	2nd Edition	ALL	Yes
2	Mathematical Elements for Computer Graphics	D. Rogers, J. Adams,	McGraw-Hill Publications	2nd Edition	ALL	Yes

4.c **List of E - Books (E – Symbol for E-Books) to be Referred by Students**

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Augmented Reality- Principles and Practice"	Dieter Schmalsteig and Tobias Hollerer	Pearson Education	2011 Edition	2-5	Yes
2	Creating Augmented and Virtual Realities- Theory and Practice for Next-Generation Spatial Computing	Erin Pangilinan, Steve Lukas, and Vasanth Mohan	O'Reilly Media	2019 Edition	2-5	NO

4.d **Reading latest / top rated research papers (at least 5 papers)**

Name of Paper	Authors with Background	Published in		Problem Statement	Available in our Library
		Date	Journal		
Revolutionizing Education Through Augmented Reality (AR) and Virtual Reality (VR): Innovations, Challenges and Future Prospects	Thangavel et al., Asian J. of Interdisciplinary Research	March 4 2025	Interdisciplinary Research https://www.journals.asianresassoc.org/index.php/ajir/article/view/3447?utm_source=chatgpt.com	This study proposes a fractal pattern modeling design method integrating computer- aided design and fractal graphics under the Internet of Things perspective,	https://journals.asianresassoc.org/index.php/ajir/article/view/3447

				enhancing fractal pattern design to a new level.	
Scoping Review of Research on Augmented Reality in Environmental Education, *Journal of Science Education & Technology*	Leonel Merino, Magdalena Schwarzl, Matthias Kraus, Michael Sedlmair,	*Apr 1 2025*.	Journal of Science Education & Technology, https://link.springer.com/article/10.1007/s10956-025-10218-z?utm_source=chatgpt.com	This study compares the effectiveness of HTML5 and OpenGL in rendering fractals on web platforms and desktop graphical programs. It evaluates their rendering capabilities and performance in graphical operations.	https://link.springer.com/article/10.1007/s10956-025-10218-z?utm_source=chatgpt.com
Advancement of Augmented Reality and Virtual Reality in Real Estate Sector	Prof. Archana TA et al	*Apr 2025*.	IJRAW https://academicjournal.ijraw.com/archive/y2025/v4/i4/mApril/Advancement%20of%20Augmented%20Reality%20and%20Virtual%20Reality%20in%20Real%20Estate%20Sector?utm_source=chatgpt.com	This paper focuses on creating a realistic model of water surfaces using height field techniques based on fractal noise. It discusses the optics of water surfaces, ray tracing, and compares GPU- and CPU-based rendering using Blender	https://academicjournal.ijraw.com/media/post/IJRAW-4-4-13.1.pdf
Exploring Immersive Virtual Reality in Higher Education: Research Gap and Future Direction – A Scoping Review	Sunardi et al.	Jan 2025	IEEE Access	This study investigates the role of data structures in diverse computing domains like web development, video games, computer graphics, AI, and operating systems. It emphasizes the necessity	https://www.researchgate.net/publication/391304862_Exploring_Immersive_Virtual_Reality_in_Higher_Education_Research_Gap_and_Future_Direction_-_A_Scoping_Review/link/6811f70360241d5140200690/download?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1YmxpY2F0aW9uIiwicGFnZSI6InB1YmxpY2F0aW9uIn19
Virtual Reality in	Nigam &	May 2025	Int. J. of	This work addresses the	https://www.research

Education	Asthana		Scientific Research	shape-color ambiguity in 3D-aware image synthesis by proposing a shading- guided generative implicit model. The model incorporates explicit illumination modeling and shading under various lighting conditions to achieve photorealistic 3D-aware image synthesis while capturing accurate underlying 3D shapes	gate.net/publication/391659647_Virtual_Reality_in_Education

4.e **Based on research paper an identify the current Problem statement**

Problem Statement	Used in						
	Quiz	Assignment	Lab	Mini Project	Poster Presentation	Test	Any Other
Use of Fractals in Cryptography and other latest Trends		Yes			Yes		

4.f **Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements**

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
Parallax Labs LLC	Yes	Yes	Yes
Feast Softwares	Yes	No	Yes
Xenium Digital Inc	Yes	Yes	Yes

4.g **Identify suitable relevant TOP Guest Speakers from Industry (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)**

Name of the Identified Guest Speaker	Designation	Name of the Company
Parvez Nasyam	Director	Xenium Digital

4.h **Identify relevant Technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]**

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event

Hack This Fall 2026 https://hackthisfall.devfolio.co/	Devfolio	18/07/2026
https://www.hackerrank.com/domains/algorithms	Hackerrank	Live Challenges

4.i **Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication**

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
MIT	Computer Graphics	Prof. Wojciech Matusik	https://academicjournal.ijraw.com/media/post/IJRAW-4-4-13.1.pdf	https://academicjournal.ijraw.com/media/post/IJRAW-4-4-13.1.pdf	Resource usage

4.j **Module Best Available in - Tick ONE best resource [from *4.a* to *4.d* in this AAP] & give details**

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Introduction and Overview of Graphics System.	T1, T3	R3	✓		✓	Y	Y
2	Raster Algorithms.	T1, T2,T3	R2				Y	Y
3	Two Dimensional Geometric Transformations, Viewing and Clipping	T1, T2,T3	R2				Y	Y
4	Three-Dimensional Object Representations, Geometric Transformations and 3D Viewing	T1, ,T,2T3	R2	✓			Y	Y
5	Visible Surface Detection	T1, T3	R2	✓			Y	Y
6	Illumination Models and Surface Rendering	T1	R2,R3	✓		✓	Y	Y

4.k **Referred to any top-rated university in that subject for content**

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
IITM	CS 6360 : Computer Graphics: Theory	Dr. Sukhendu Das	Recorded	https://www.cse.iitm.ac.in/~vplab/computer_graphics.html

	and Algorithms			
MIT OCW	Computer Graphics	Prof. Wojciech Matusik/ And Prof. Frédo Durand	Recorded	https://ocw.mit.edu/courses/6-837-computer-graphics-

4.l **Faculty received any certification related to this subject. List of Certifications Identified / Done**

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Computer Graphics with modern openGL and C++	Udemy	22 Hours		OpenGL	--	Dec., 2026	--

4.m **Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done**

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	Udemy	2025		Presentations with ChatGPT
PBL				
Sub. Content Training				

4.n **Best Practices Identified and adopted**

No.	Item	Best Practices Identified		
		Univ. 1	Univ. 2	Univ. 3
1	Microsite	IBM	Boston University	Carnegie Mellon University
2	Video Lectures	MIT	University of Illinois	Berkeley university
3	Assignments			
4	Mini Project			
5	Assessment Metric			

6	Quizzes	University of Illinois	Berkeley university	Trinity, Dublin
7	Labs/ Practical (PBL)			
8	Tests	University of Illinois	Berkeley university	Trinity, Dublin
9	Etc			
10	Peer Assessment etc.			

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	NPTEL Video on Introduction to Computer Graphics by Dr. Sukendu Das (IITC) https://www.youtube.com/watch?v=fwzYuhduME NPTEL NOC: Virtual Reality Engineering, IIT Madras https://nptel.ac.in/courses/121106013 NPTEL : Virtual Reality, IIT Madras https://nptel.ac.in/courses/106106138 Demo on Augmented Reality https://www.youtube.com/watch?v=oH_LfXnklRw Augmented Reality and its Applications https://www.youtube.com/watch?v=tSc4YcuFG50	1
2	NPTEL Video on CRT Display Devices by Dr. Sukendu Das (IITC) https://www.youtube.com/watch?v=0ZuSu44-WeE https://www.youtube.com/watch?v=myRKFWMNKO0&index=3&list=PL338D19C40D6D1732 https://www.youtube.com/watch?v=GzpaDTbXMiQ&list=PL338D19C40D6D1732&index=4 https://www.youtube.com/watch?v=CoBlpAZQYiA&index=5&list=PL338D19C40D6D1732	1
3	NPTEL Video on Scan Converting Lines , Circles and Ellipses by Dr. Sukendu Das (IITC) https://www.youtube.com/watch?v=E_sqVKiuCx8&index=14&list=PL338D19C40D6D1732 https://www.youtube.com/watch?v=frrBDwmkQ0E&index=15&list=PL338D19C40D6D1732 https://www.youtube.com/watch?v=_f_pHgRQPPE&list=PL338D19C40D6D1732&index=16 https://www.youtube.com/watch?v=FO5XLAMh9E0&index=17&list=PL338D19C40D6D1732 https://www.youtube.com/watch?v=TNbkX5bYrtE&index=18&list=PL338D19C40D6D1732	2
4	NPTEL Video on Transformations in 2D by Dr. Sukendu Das (IITC) https://www.youtube.com/watch?v=DPcVMEBDpAY https://www.youtube.com/watch?v=VziP-jctzns	3
5	NPTEL Video on Clipping by Dr. Prem Kalra (IITD) https://www.youtube.com/watch?v=7iYggzuo6Ks https://www.youtube.com/watch?v=lZJI_Fth66A&list=PL112A527F83F7A5E4&index=5	4

6	NPTEL Video on 3D Viewing by Dr. Prem Kalra (IITD) https://www.youtube.com/watch?v=k2ajK9uM0Co&list=PL112A527F83F7A5E4&index=8 https://www.youtube.com/watch?v=el2s6VIeyyQ&list=PL112A527F83F7A5E4&index=9	5
7	NPTEL Video on Solid Modeling by Dr. Sukendu Das (IITC) https://www.youtube.com/watch?v=uDlsWxDA1TQ https://www.youtube.com/watch?v=LZEVG3raveE	5

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	3D Data Visualization for Science Communication (Coursera)	Coursera	4 weeks	Y
2	Interactive Computer Graphics (Coursera)	Coursera	4 weeks	Y
	https://www.udemy.com/course/graphics-with-modern-opengl/?couponCode=MT260714G2	Ben Cook	22 Hrs	Y

5. Consolidated Course Lesson Plan

	From (date/month/year)	To (date/month/year)	Total Number of Weeks
Semester Duration	6th July 2026	23rd Oct. 2026	15

Week	Lecture no.	Module No.	Lecture Topics / IA 1 and IA 2 / BSA planned to be covered	Actual date of Completion	COs	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lect No. (on LMS)	Chapter No. / Page Nos./ Books/ Web Site
1	1	1	Definition and Representative uses of computer graphics, classification of application areas, Overview of coordinate systems ,definition of scan conversion, rasterization and rendering. Raster scan & random scan displays, Flat Panel displays like LCD and LED , architecture of raster graphics system with display processor, architecture of random scan systems.		CO1	Understand	--	--
2	2	2	Scan conversions of point, line, circle and ellipse : DDA algorithm		CO2	Understand and Apply	--	T1 Ch1 (Pg 4- 27)
3	3	2	Bresenham algorithm for line drawing, midpoint algorithm for circle		CO2	Analyze Apply		
4	4	2	midpoint algorithm for ellipse drawing and Numericals		CO2	Understand Apply		
Week	Lecture no.	Module No.	Lecture Topics / IA 1 and IA 2 / BSA planned to be covered	Actual date of Completion	COs	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lect No. (on LMS)	Chapter No. / Page Nos./ Books/ Web Site
5	5	2	Filled Area Primitive: Scan line Polygon Fill algorithm, Inside outside tests		CO2	Analyze Apply		
6	6	2	Boundary Fill and Flood fill algorithm		CO2	Analyze Apply		T1 Ch2 (Pg 34-78)
7			MSE					

8	8	3	Basic transformations : Translation , Scaling , Rotation •Matrix representation and Homogeneous Coordinates •Composite transformation		CO3	Apply	--	
9	9	3	•Other transformations : Reflection and Shear •Raster method for transformation.		CO3	Analyze Apply		T1 Ch4 (Pg 123-170)
10	10	4	Two Dimensional Viewing and Clipping •Viewing transformation pipeline and Window to Viewport coordinate transformation		CO4	Analyze Apply		
11	11	4	•Clipping operations – Point clipping , Line clipping algorithms : Cohen – Sutherland Midpoint subdivision , Liang – Barsky , Polygon Clipping Algorithms : Sutherland – Hodgeman, Weiler – Atherton.		CO4	Analyze Apply		
12	12	5	3D Transformations : Translation, Rotation , Scaling and Reflection		CO5	Apply	--	--
13	13	5	Composite transformations :Rotation about an arbitrary axis 3D transformation pipeline		CO5	Understand		T1 Ch8(Pg 272-306)
14	14	5	Projections – Parallel , Perspective.(Matrix Representation) 3D clipping.		CO5	Apply		
15	15	6	Classification of Visible Surface Detection algorithm: Back Surface detection method, Depth Buffer method, Depth Sorting method, Scan line method, Area Subdivision method		CO6	Understand		T1 Ch9(Pg 312-325)

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA=15M)							
Class Participation	Assignments	Video Lecture summary	Seminar	Poster Presentation	POP Quiz +Crossword	Paper Summary	Total
5	05	01	01	01	01	01	15

Practical (ISA=25M)					Total

Class Participation	Lab-1	Lab -2	A Lab -3	Lab -4	Lab -5	Lab -6	Lab -7	Lab -8	Lab -9	Lab -10		
5	2	2	2	2	2	2	2	2	2	2		25

Class Participation	MSE-1	MSE-2	ESE	Total
15	20	20	40	75

7. Assignments / Tutorials Details (must attach print out of all questions together with AAP)

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
1	Poster/ Introduction Computer Graphics	C01	Create	06/07/2026	27/7/2026
2	Bresenham Line/ Scan conversion algorithms	CO2	Apply	06/07/2026	03/08/2026
3	Midpoint Circle/ Two Dimensional viewing and clipping	CO2	Apply	06/07/2026	10/08/2026
4	Area Filling/ Three Dimensional Geometric Transformations and 3D Viewing	CO2	Apply	06/07/2026	17/08/2026
5	2D Transformations/ Visible Surface Detection & Illumination Models	CO3	Apply	06/07/2026	24/08/2026
6	Window To Viewport	CO3	Apply	06/07/2026	03/08/2026
7	Line Clipping	CO4	Apply	06/07/2026	10/08/2026
8	Polygon Clipping	CO4	Apply	06/07/2026	17/08/2026
9	Surface Detection	CO5	Apply	06/07/2026	24/08/2026

10	Light and Shading	CO5	Understand	06/07/2026	07/09/2026
11	Mind-Map	CO1 to CO6	Create	06/07/2026	15/09/2026

* Tick (√) the Type of the Assignment: Regular (R); Pop Quiz (PQ) ; Open Book Test for TE/BE/ME (OBT)
Write number for text book, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
MSE-1	5th Week	1,2	CO1-CO2	Q.1. Solve 2 out of 3 (10 Marks) Q.2. Solve 1 out of 2 (10 Marks)	No re-examination for MSE
MSE-2	10th Week	3,4	CO3	Q.1. Solve 1 out of 2 (10 Marks) Q.2. Solve 1 out of 2 (10 Marks)	No re-examination for MSE
Pop Quiz	20/09/2026	ALL	CO2-CO5	NA	NA
Croswword	24/09/2026	ALL	CO1-CO6	NA	NA

9. Practical Activities – ALL NEW

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	2	Implement DDA Line Drawing algorithms and Bresenham algorithm.		√	Raster Algorithms	CO2
2	2	Program to display thick line, dotted line, and dashed line.		√	Raster Algorithms	CO2
3	2	Implement Area Filling Algorithm: Boundary Fill, Flood Fill, Scan line Polygon Fill		√	Raster Algorithms	CO2
4	2	Implement Curve: Bezier for n control points		√	Curve Generation	CO2
5	3	Implement 2D Transformations: Translation, Scaling, Rotation, Reflection, Shear.		√	2D Transformations	CO3
6	3	Implement Line Clipping Algorithm: Cohen Sutherland / Liang Barsky.		√	Line Clipping	CO3
7	3	Implement polygon clipping algorithm.		√	Polygon Clipping	CO3
8	4	Program to represent a 3D object using polygon surfaces and then perform 3D transformation		√	3D Modelling	CO4

9	4	Program to perform projection of a 3D object on Projection Plane: Parallel and Perspective		√	Projection	CO4
10	4	PBLE :Study of Open GL library functions and using it to generate graphical objects	√		Rendering	CO4
11	6	PBLE: Using Unity Software generate Animation with light and shading effect	√		Rendering	CO5, CO6

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	Experiential learning/Interaction with Outside World	1- Guest Lectures by Industry Expert		Guest lecture of industry expert based on "Animation and Special Effects"
		2- Workshops		Yes
		3- Mini Project		Yes
		4- Industrial Visit		--
		5- Any other activity		--
2	Collaborative & Group Activity	6- Poster Presentation		Individual students need to prepare posters based on Mind Map
		7- Minute Papers		Minute paper activity will be conducted on specific topic or on entire syllabus
		8- Students Seminars		Yes
		9- Students Debates		--
		10- Panel Discussion / Mock GD		--
		11- Mock Interview		--
		12- Any other activity		--
3	Co-Curricular Activity	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)		NPTEL/ YouTube videos on a specific topic will be shown to students based on 4.k How Virtual Reality Works https://www.edx.org/course/how-virtual-reality-works Introduction to XR: VR, AR, and MR Foundations
		14- Lecture Capture Usage		All lecture recordings are made available to students
		15- Any other activity		--

4	**Tests & Assessments**	16- Class Tests/ Weekly Tests		--
		17- Pop Quiz		Yes
		18- Mobile App Based Quiz		--
		19- Open Book Test		Yes
		20- Take Home Test		--
		21- Any other activity		--

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	TE –SEM 5	Computer Graphics	Yes	14th July 2026

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	TE –SEM 5	Computer Graphics	Yes	14th July 2026

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Dr. Santosh Tamboli

External Industry Mentor	External Academic Mentor	VIT Cluster Mentor	Program HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	Assessment Criteria: The Learner can:	Evaluated under ISA/MSE/ESE/LAB
LO1.1 Explain the evolution, applications, and importance of computer graphics in various domains. LO1.2 Describe different coordinate systems used in computer graphics. LO1.3 Differentiate between raster scan and random scan display systems. LO1.4 Explain the architecture of raster graphics systems and display processors. LO1.5 Illustrate graphics primitives and their representation in computer graphics. LO1.6 Explain rendering and rasterization processes used in graphics systems.	• Explain the evolution, scope, and applications of computer graphics in engineering, entertainment, CAD/CAM, medical imaging, and scientific visualization. • Distinguish different coordinate systems and demonstrate their use in graphical applications. • Compare raster scan and random scan display systems based on architecture, advantages, limitations, and applications. • Describe the working principles of raster graphics systems, display processors, frame buffers, and display devices. • Identify and represent graphics primitives such as points, lines, circles, polygons, and curves. • Explain rendering and rasterization processes involved in displaying graphical objects.	ISA/MSE/ESE
LO2.1 Implement DDA algorithm for line drawing. LO2.2 Implement Bresenham's line drawing algorithm. LO2.3 Implement Midpoint and Bresenham's circle drawing algorithms. LO2.4 Compare aliasing and anti-aliasing techniques. LO2.5 Apply Boundary Fill and Flood Fill algorithms. LO2.6 Implement Scan Line polygon filling algorithm.	• Implement DDA algorithm for drawing straight lines accurately. • Develop Bresenham's algorithms for efficient line and circle generation. • Compare DDA and Bresenham's algorithms based on computational efficiency and accuracy. • Implement Boundary Fill, Flood Fill, and Scan Line polygon filling techniques. • Analyze the effects of aliasing and apply anti-aliasing techniques to improve image quality. • Evaluate scan conversion algorithms for different graphical objects.	ISA/MSE/ESE/LAB
LO3.1 Apply translation, scaling, and rotation transformations. LO3.2 Perform composite transformations using homogeneous coordinates. LO3.3 Apply 3D geometric transformations. LO3.4 Perform rotation about an arbitrary axis. LO3.5 Apply parallel and perspective projections. LO3.6 Implement transformation matrices in graphics applications.	• Apply 2D transformation matrices for translation, scaling, rotation, reflection, and shearing. • Implement composite transformations using homogeneous coordinates. • Apply 3D transformations including translation, scaling, rotation, reflection, and projection. • Perform arbitrary axis rotation using appropriate transformation matrices. • Differentiate parallel and perspective projection techniques for graphical applications. • Develop programs implementing	ISA/MSE/ESE/LAB

	transformation operations using OpenGL or equivalent graphics libraries.	
LO4.1 Explain visible surface detection methods. LO4.2 Compare Depth Buffer, Scan Line, and Area Subdivision methods. LO4.3 Analyze hidden surface removal algorithms. LO4.4 Explain illumination models. LO4.5 Implement Constant, Gouraud, and Phong shading techniques. LO4.6 Evaluate rendering quality using different shading methods.	• Explain the need for hidden surface removal in 3D graphics. • Compare different visible surface detection algorithms based on efficiency and applications. • Analyze illumination models including ambient, diffuse, and specular reflection. • Implement Constant, Gouraud, and Phong shading techniques using graphics programming. • Evaluate rendered images based on lighting and shading quality. • Recommend suitable hidden surface removal and shading techniques for different graphical applications.	ISA/MSE/ESE/LAB
LO5.1 Explain viewing transformation pipeline. LO5.2 Apply window-to-viewport transformation. LO5.3 Implement Cohen–Sutherland line clipping. LO5.4 Apply Liang–Barsky clipping algorithm. LO5.5 Implement Sutherland–Hodgeman polygon clipping. LO5.6 Analyze clipping techniques.	• Explain the stages involved in the viewing transformation pipeline. • Perform window-to-viewport transformation for given graphical objects. • Implement Cohen–Sutherland and Liang–Barsky line clipping algorithms. • Implement Sutherland–Hodgeman polygon clipping algorithm. • Compare clipping algorithms based on computational efficiency and practical applications. • Develop OpenGL programs demonstrating viewing and clipping operations.	ISA/MSE/ESE/LAB
LO6.1 Generate Bezier curves using control points. LO6.2 Construct B-Spline curves. LO6.3 Generate fractal curves such as Koch curves. LO6.4 Design graphical models using curve representation techniques. LO6.5 Evaluate different curve generation techniques. LO6.6 Develop graphics applications integrating curves, transformations, and rendering.	• Generate Bezier and B-Spline curves using suitable mathematical models. • Construct fractal curves and analyze their graphical characteristics. • Design graphical objects using curve representation techniques. • Compare Bezier, B-Spline, and fractal curves based on continuity, flexibility, and applications. • Develop integrated graphics applications using curves, transformations, and rendering techniques. • Evaluate graphical models for accuracy, smoothness, rendering quality, and computational efficiency.	ISA/MSE/ESE/LAB